BEU CONNECT

B.Tech 6th Semester Exam., 2022

(New Course)

Time: 3 Hours

Code: 100602 *Subject: Computer Network* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer from the following (any seven): **[2×7=14]**

(a) A proxy server is used as the computer
- (i) with external access
- (ii) acting as a backup
- (iii) performing file handling
- (iv) accessing user permissions

(b) The term 'WAN' stands for
- (i) Wide Area Net
- (ii) Wide Access Network
- (iii) Wide Area Network
- (iv) Wide Access Net

(c) Which layer of the TCP/IP stack corresponds to the OSI model transport layer?
- (i) Host-to-host
- (ii) Application
- (iii) Internet
- (iv) Network access

(d) What is the maximum header size of an IP packet?
- (i) 32 bytes
- (ii) 64 bytes
- (iii) 30 bytes
- (iv) 60 bytes

(e) The address which is used to identify a process on a host is
- (i) physical address
- (ii) port address
- (iii) logical address
- (iv) specific address

(f) Which sublayer of the data link layer performs circuit functions that depend on the kind of medium?
- (i) Media access control sublayer
- (ii) Logical link control sublayer
- (iii) Network interface control sublayer
- (iv) Both (i) and (ii)

(g) Transport layer protocol deals with
- (i) application-to-application communication
- (ii) node-to-node communication
- (iii) the process-to-process communication
- (iv) Both (i) and (iii)

(h) FTP server
- (i) maintains state information
- (ii) is stateless
- (iii) has single TCP connection for a file transfer
- (iv) has UDP connection for file transfer

(i) Which of the following is the multiple access protocol for channel access control?
- (i) CSMA/CD
- (ii) CSMA/CA
- (iii) Both CSMA/CD and CSMA/CA
- (iv) HDLC

(j) Physical layer provides
- (i) mechanical specifications of electrical connectors and cables
- (ii) electrical specification of transmission line signal level
- (iii) specification for IR over optical fiber
- (iv) All of the above

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Q.2 Explain the duties of different layers in TCP/IP model. **[14]**

Q.3 What do you mean by the term 'ethernet'? What are the types of cable commonly used for ethernet? **[14]**

Q.4 Differentiate between pure aloha and slotted aloha with examples. **[14]**

Q.5 How data link layers provide flow and error control? Explain with examples. **[14]**

Q.6 Describe RIP routing protocol with an example. Explain count to infinity problem and its countermeasures. **[14]**

Q.7 An ISP is granted the block 80.70.56.0/21. The ISP needs to allocate addresses for two organizations each with 500 addresses, two organizations each with 250 addresses, and three organizations each with 50 addresses. **[14]**

(a) Find the number and range of addresses in the ISP block.

(b) Find the range of addresses for each organization and the range of unallocated addresses.

Q.8 How transport layer provides congestion control? Differentiate between leaky bucket and token bucket algorithms. **[14]**

Q.9 Explain the difference between TCP and UDP. A client residing on a host with IP address 122.45.12.7 sends a message to the corresponding server residing on a host with IP address 200.112.45.90. If the well-known port is 161 and the ephemeral port is 51000, what are the pair of socket addresses used in this communication? **[14]**

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